



Ext 2

NORTH SYDNEY BOYS HIGH SCHOOL

2014
ASSESSMENT TASK 2

Mathematics

Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 55 minutes
- A table of standard integrals is provided
- Write using blue or black pen
- Board approved calculators may be used
- Questions 1 – 4 to be answered on multiple choice answer sheet provided
- All necessary working should be shown for Questions 5 – 10 in the answer booklet provided
- Each new question is to be started on a **new page**.

Class Teacher:

(Please tick or highlight)

- ☐ Ms Ziazaris
- ☐ Mr Lam
- ☐ Mr Ireland

Student Number:

(To be used by the exam markers only.)

Question No	1-4	5	6	7	8	9	10	Total	Total
Mark	$\overline{4}$	$\overline{7}$	$\overline{4}$	$\overline{5}$	$\overline{13}$	$\overline{6}$	$\overline{4}$	$\overline{43}$	$\overline{100}$

Questions 1-4 - Answer on the multiple choice answer sheet provided.

1. If $\frac{x^2}{48} + \frac{y^2}{12} = 1$ is the equation of an ellipse, then the equations of the directrices are :-

(A) $x = \pm 2$

(C) $x = \pm 4$

(B) $x = \pm 8$

(D) $x = \pm \frac{8\sqrt{3}}{\sqrt{5}}$

2. If $\frac{y^2}{16} - \frac{x^2}{25} = 1$ is the equation of a hyperbola, then its eccentricity, e is:-

(A) $e = \frac{\sqrt{41}}{4}$

(C) $e = \sqrt{41}$

(B) $e = \frac{\sqrt{41}}{5}$

(D) $e = \frac{5\sqrt{41}}{4}$

3. $\int 3 \cos \frac{1}{3} x dx =$

(A) $\sin \frac{1}{3} x + C$

(C) $-9 \sin \frac{1}{3} x + C$

(B) $-\sin \frac{1}{3} x + C$

(D) $9 \sin \frac{1}{3} x + C$

4. On a hyperbola, the distance between its vertices is 8 units and the distance between its foci is 10 units. The acute angle between its asymptotes will be

(A) $36^\circ 52'$

(C) 45°

(B) $73^\circ 44'$

(D) $75^\circ 34'$

CONTINUED...

Questions 5-10 – Answer in the booklet provided.

5. (a) $\int (x+1) \sec^2(x^2 + 2x) dx$ 2
- (b) $\int \sin^2 3x dx$ 2
- (c) The region under the curve $y = \tan x$ between $x = \frac{\pi}{6}$ and $x = \frac{\pi}{3}$ is rotated about the x-axis. What is the volume of the solid of revolution formed. 3
6. Find the equations of the two tangents to the ellipse $16x^2 + 25y^2 = 400$ which are parallel to the line $y = x + 2$. 4
7. (a) Write $\frac{4x^2 - 5x - 7}{(x-1)(x^2 + x + 2)}$ in the form $\frac{A}{x-1} + \frac{Bx+C}{x^2 + x + 2}$. 3
- (b) Hence, evaluate $\int \frac{4x^2 - 5x - 7}{(x-1)(x^2 + x + 2)} dx$ 2

CONTINUED...

8. The hyperbola $16x^2 - 9y^2 = 144$ has foci $S(5,0)$ and $S'(-5,0)$.

The directrices are $x = \frac{9}{5}$ and $x = -\frac{9}{5}$.

- (a) State the equation of each asymptote of the hyperbola. 1
- (b) Sketch the hyperbola and indicate on your diagram the foci, directrices, and asymptotes. 2
- (c) By differentiation, find the gradient of the tangent to the hyperbola at $P(3 \sec \theta, 4 \tan \theta)$. 2
- (d) Show that the tangent to the hyperbola at P has equation $4x = (3 \sin \theta)y + 12 \cos \theta$. 2
- (e) Given that $0 < \theta < \frac{\pi}{2}$, show that Q , the point of intersection of the tangent to the hyperbola at P and the nearer directrix, has y coordinate $\frac{12 - 20 \cos \theta}{5 \sin \theta}$. 2
- (f) Calculate the gradients of SP and SQ . 2
- (g) Determine whether $\angle PSQ$ is a right angle. 2

9. Consider the equation $\frac{x^2}{3-\lambda} + \frac{y^2}{5-\lambda} = 1$

- (a) Determine the real values for which the above equation defines an
 - (i) Ellipse 1
 - (ii) Hyperbola 2
- (b) Sketch the curve for $\lambda = 4$, showing the branches and vertices only. 1
- (c) Describe how the shape of the curve changes as λ increases from 3 towards 5. 1
- (d) What is the limiting position of the curve as λ approaches 5. 1

10. Prove by mathematical induction that $\sqrt{2 + \sqrt{2 + \sqrt{2 + \dots \sqrt{2}}}} = 2 \cos \frac{\pi}{2^{n+1}}$ for $n \geq 1$

where there are n lots of 2 's on the left hand side. 4

$$\left(\text{eg. } \sqrt{2 + \sqrt{2}} = 2 \cos \frac{\pi}{2^{2+1}} \right)$$

END OF EXAMINATION

Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g “●”

STUDENT NUMBER:

Class (please ✓)

☐ 12M4A – Ms Ziazaris

☐ 12M4B – Mr Lam

☐ 12M4C – Mr Ireland

1 – ☐ A ☐ B ☐ C ☐ D

2 – ☐ A ☐ B ☐ C ☐ D

3 – ☐ A ☐ B ☐ C ☐ D

4 – ☐ A ☐ B ☐ C ☐ D

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1} + C, \quad n \neq -1; \quad x \neq 0 \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x + C, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax + C, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax + C, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax + C, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax + C, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C, \quad a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right) + C, \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right) + C$$

NOTE: $\ln x = \log_e x$, $x > 0$

EXTENSION 2 - 2014 HSC ASSESSMENT TASK 2 SOLN'S

$$\begin{aligned} \textcircled{1} \quad a^2 &= 48 & e^2 &= 1 - \frac{b^2}{a^2} \\ a &= 4\sqrt{3} & e^2 &= 1 - \frac{12}{48} \\ b^2 &= 12 & e &= \frac{\sqrt{3}}{2} \\ b &= 2\sqrt{3} \end{aligned}$$

Dir : $x = \pm \frac{a}{e}$

$$x = \pm \frac{4\sqrt{3}}{\frac{\sqrt{3}}{2}}$$

$$\boxed{x = \pm 8 \text{ --- B}}$$

$$\textcircled{2} \quad e^2 = 1 + \frac{a^2}{b^2}$$

$$e^2 = 1 + \frac{25}{16}$$

$$\boxed{e = \frac{\sqrt{41}}{4} \text{ --- A}}$$

$$\textcircled{3} \quad \int 3 \cos \frac{1}{3}x \, dx$$

$$= 3 \int \cos \frac{1}{3}x \, dx$$

$$= 3 \times 3 \sin \frac{1}{3}x + C$$

$$= \boxed{9 \sin \frac{1}{3}x + C \text{ --- D}}$$

$$\textcircled{4} \quad a = 4$$

$$ae = 5$$

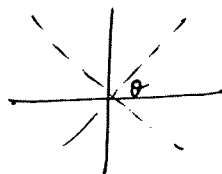
$$e = \frac{5}{4}$$

$$e^2 = 1 + \frac{b^2}{a^2}$$

$$\therefore b = 3$$

Asymptotes :

$$y = \pm \frac{b}{a}x$$



$$\tan \theta = \frac{b}{a}$$

$$\tan \theta = \frac{3}{4} \quad \therefore \theta = 36^\circ 52'$$

$$\boxed{\text{Angle between asymptotes } 73^\circ 44' \text{ --- B}}$$

$$5. a) \int (x+1) \sec^2(x^2+2x) dx$$

$$= \frac{1}{2} \int (2x+2) \sec^2(x^2+2x) dx$$

$$= \frac{1}{2} \frac{\sec^3(x^2+2x)}{3} + C$$

$$= \frac{\sec^3(x^2+2x)}{6} + C$$

~ 1 mark

~ 1 mark

$$b) \int \sin^2 3x dx$$

$$= \frac{1}{2} \int (1 - \cos 6x) dx$$

$$= \frac{1}{2} \left[x - \frac{\sin 6x}{6} \right] + C$$

$$= \frac{x}{2} - \frac{\sin 6x}{12} + C$$

$$\cos 2x = 1 - 2\sin^2 x$$

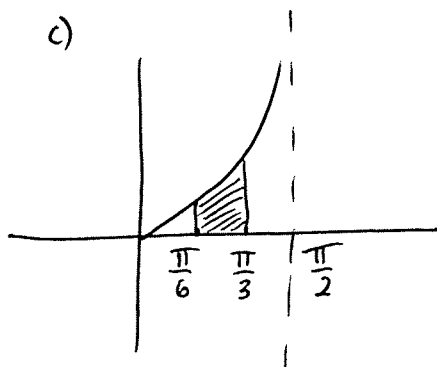
$$\sin^2 x = \frac{1 - \cos 2x}{2}$$

$$\sin^2 3x = \frac{1 - \cos 6x}{2}$$

~ 1 mark to convert to $\cos 6x$

~ 1 mark to integrate

c)



$$V = \pi \int_{\pi/6}^{\pi/3} \tan^2 x dx$$

$$= \pi \int_{\pi/6}^{\pi/3} (\sec^2 x - 1) dx$$

$$= \pi \left[\tan x - x \right]_{\pi/6}^{\pi/3}$$

$$= \pi \left[\tan \frac{\pi}{3} - \frac{\pi}{3} - \tan \frac{\pi}{6} + \frac{\pi}{6} \right]$$

$$= \pi \left[\sqrt{3} - \frac{1}{\sqrt{3}} - \frac{\pi}{6} \right] \text{ c. units.}$$

1 mark to set up integral

1 mark to integrate

1 mark for answer
(unsimplified is ok)

6. Let the tangents have eqn

$$y = mx + c$$

$$\text{But } m = 1$$

$$\therefore y = x + c \quad \text{--- (1)}$$

$$16x^2 + 25y^2 = 400 \quad \text{--- (2)}$$

1 mark

Sub (1) into (2)

$$16x^2 + 25(x+c)^2 = 400$$

$$16x^2 + 25x^2 + 50xc + 25c^2 = 400$$

$$41x^2 + 50xc + 25c^2 - 400 = 0$$

For tangency, $\Delta = 0$.

1 mark
for tangency

$$\text{ie. } (50c)^2 - 4(41)(25c^2 - 400) = 0$$

$$2500c^2 - 164(25c^2 - 400) = 0$$

$$2500c^2 - 4100c^2 + 65600 = 0$$

$$-1600c^2 = -65600$$

$$c^2 = 41$$

$$c = \pm\sqrt{41}$$

1 mark

$\therefore$ Eqn's of tangents are $y = x \pm \sqrt{41}$

1 mark.

$$\begin{aligned}
 7. a) \quad \frac{4x^2 - 5x - 7}{(x-1)(x^2+x+2)} &= \frac{A}{x-1} + \frac{Bx+C}{x^2+x+2} \\
 &= \frac{A(x^2+x+2) + (Bx+C)(x-1)}{(x-1)(x^2+x+2)} \\
 &= \frac{Ax^2 + Ax + 2A + Bx^2 - Bx + Cx - C}{(x-1)(x^2+x+2)} \\
 &= \frac{(A+B)x^2 + (A-B+C)x + 2A - C}{(x-1)(x^2+x+2)}
 \end{aligned}$$

1 mark

$$\therefore A+B=4 \quad \text{--- (1)}$$

$$A-B+C = -5 \quad \text{--- (2)}$$

$$2A-C = -7 \quad \text{--- (3)}$$

Sub $B = 4-A$ into (2)

$$A - (4-A) + C = -5$$

$$A - 4 + A + C = -5$$

$$2A + C = -1 \quad \text{--- (4)}$$

$$(3) + (4)$$

$$4A = -8$$

$$A = -2$$

$$\therefore C = -1 - 2(-2)$$

$$C = 3$$

$$B = 6$$

1 mark getting to A.

} 1 mark for B, C

$$\therefore \frac{4x^2 - 5x - 7}{(x-1)(x^2+x+2)} = \frac{-2}{x-1} + \frac{6x+3}{x^2+x+2}$$

$$b) \int \left(\frac{-2}{x-1} + \frac{6x+3}{x^2+x+2} \right) dx$$

$$= -2 \ln|x-1| + 3 \int \frac{2x+1}{x^2+x+2} dx$$

1 mark

$$= -2 \ln|x-1| + 3 \ln|x^2+x+2| + C$$

1 mark

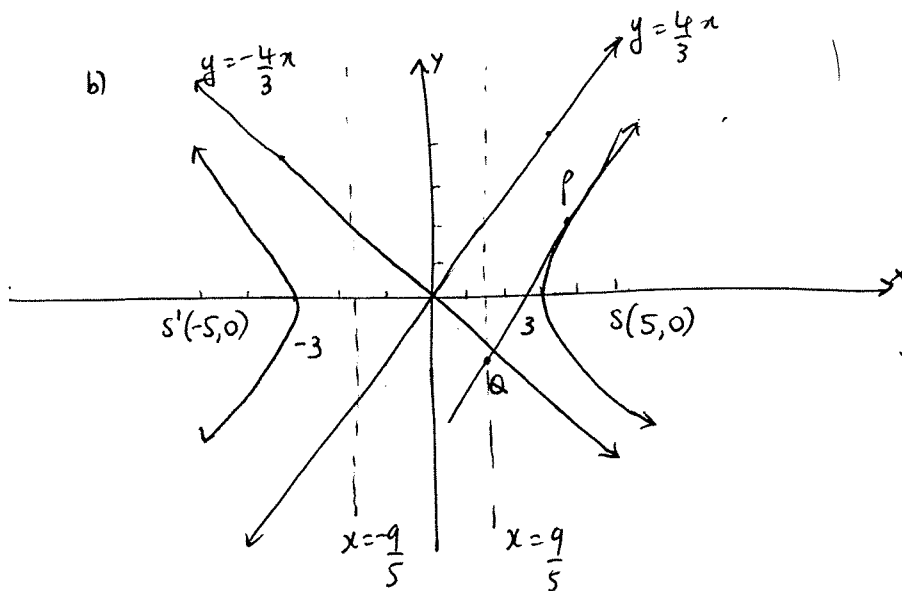
8. $16x^2 - 9y^2 = 144$

$$\frac{x^2}{9} - \frac{y^2}{16} = 1$$

a) $y = \pm \frac{b}{a}x$

$$y = \pm \frac{4}{3}x$$

— 1 mark



2 { — 1 mark for vertices & foci
— 1 mark for directrices/asympt.

c) $\frac{2x}{9} - \frac{2y}{16} \frac{dy}{dx} = 0$

$$\frac{dy}{dx} = \frac{2x}{9} \times \frac{16}{2y}$$

$$\frac{dy}{dx} = \frac{16x}{9y}$$

At $(3\sec\theta, 4\tan\theta)$

$$\frac{dy}{dx} = \frac{4\sec\theta}{3\tan\theta} = \frac{4}{3\sin\theta}$$

2 { — 1 mark for $\frac{dy}{dx}$
— 1 mark for subⁿ of pt.

d) Equ'n of tangent

$$y - 4\tan\theta = \frac{4}{3\sin\theta} (x - 3\sec\theta)$$

— 1 mark for subⁿ

$$3y\sin\theta - 12\tan\theta\sin\theta = 4x - 12\sec\theta$$

$$3y\sin\theta - 12\tan\theta\sin\theta + \frac{12}{\cos\theta} = 4x$$

$$3y\sin\theta + \frac{12 - 12\sin^2\theta}{\cos\theta} = 4x$$

$$3y\sin\theta + 12\cos\theta = 4x$$

— 1 mark for answer

$$e) 4x = 3y \sin \theta + 12 \cos \theta \quad \text{--- ①}$$

$$x = \frac{9}{5} \quad \text{--- ②}$$

Solve simult. to obtain Q.

Sub ② into ①

$$4\left(\frac{9}{5}\right) = 3y \sin \theta + 12 \cos \theta$$

_____ 1 mark for subⁿ

$$\frac{36}{5} - 12 \cos \theta = 3y \sin \theta$$

$$y = \frac{\frac{36}{5} - 12 \cos \theta}{3 \sin \theta}$$

$$y = \frac{36 - 60 \cos \theta}{15 \sin \theta}$$

$$y = \frac{12 - 20 \cos \theta}{5 \sin \theta}$$

_____ 1 mark for answer

$$f) m_{SP} = \frac{4 \tan \theta}{\sec \theta - 5}$$

_____ 1 mark for m_{SP}

$$m_{SQ} = \frac{\frac{12 - 20 \cos \theta}{5 \sin \theta}}{\frac{9}{5} - 5}$$

$$= \frac{12 - 20 \cos \theta}{-16 \sin \theta}$$

$$= \frac{5 \cos \theta - 3}{4 \sin \theta}$$

_____ 1 mark for m_{SQ}

$$g) m_{SP} \times m_{SQ} = \frac{4 \tan \theta}{3 \sec \theta - 5} \times \frac{5 \cos \theta - 3}{4 \sin \theta}$$

$$= \frac{5 \tan \theta \cos \theta - 3 \tan \theta}{\sin \theta (3 \sec \theta - 5)}$$

$$= \frac{5 \sin \theta - \frac{3 \sin \theta}{\cos \theta}}{\left(\frac{3}{\cos \theta} - 5\right) \sin \theta}$$

$$= \frac{5 \sin \theta \cos \theta - 3 \sin \theta}{(3 - 5 \cos \theta) \sin \theta}$$

$$= \frac{\sin \theta (5 \cos \theta - 3)}{(3 - 5 \cos \theta) \sin \theta} = -1$$

_____ 1 mark for changing to sin/cos.

_____ 1 mark for obtaining -1

Since $m_{SP} \times m_{SQ} = -1$ then $\angle PSQ = 90^\circ$.

9. $\frac{x^2}{3-\lambda} + \frac{y^2}{5-\lambda} = 1$

a) (i) Ellipse

$$3-\lambda > 0 \text{ and } 5-\lambda > 0$$

$$\text{ie. } \lambda < 3 \text{ and } \lambda < 5$$

$$\therefore \boxed{\lambda < 3}$$

(ii) Hyperbola

$$3-\lambda > 0 \text{ and } 5-\lambda < 0$$

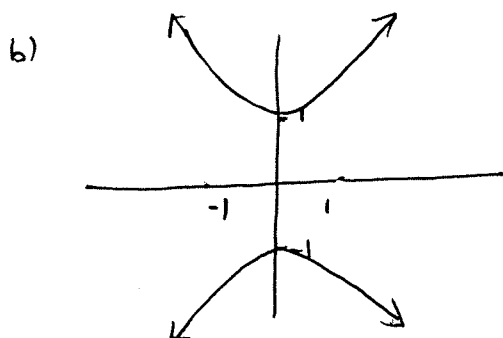
$$\text{ie. } \lambda < 3 \text{ and } \lambda > 5$$

not possible.

$$\text{OR. } 3-\lambda < 0 \text{ and } 5-\lambda > 0$$

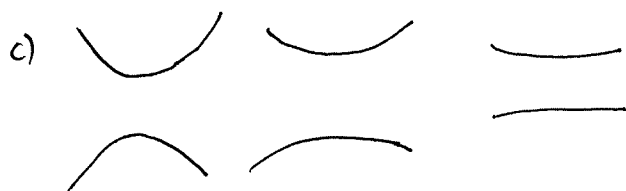
$$\text{ie. } \lambda > 3 \text{ and ie. } \lambda < 5$$

$$\therefore \boxed{3 < \lambda < 5}$$



$$\frac{x^2}{-1} + \frac{y^2}{1} = 1$$

$$\text{ie. } y^2 - x^2 = 1$$



The branches become flatter almost parallel lines. ie. Concavity of branches decreases.

d) Curve approaches 2 parallel lines which converge to $y=0$.

10. $\sqrt{2+\sqrt{2+\sqrt{2+\dots\sqrt{2}}}} = 2 \cos \frac{\pi}{2^{n+1}}$ for $n \geq 1$

Step 1: Show true for $n=1$

$$\text{LHS} = \sqrt{2}$$

$$\text{RHS} = 2 \cos \frac{\pi}{2^{1+1}}$$

$$= 2 \cos \frac{\pi}{4}$$

$$= 2 \times \frac{1}{\sqrt{2}}$$

$$= \sqrt{2}$$

LHS = RHS $\therefore$ True for $n=1$

Step 2: Assume true for $n=k$

ie. $\underbrace{\sqrt{2+\sqrt{2+\sqrt{2+\dots\sqrt{2}}}}}_{k \text{ 2's}} = 2 \cos \frac{\pi}{2^{k+1}}$

Step 3: Prove true for $n=k+1$

$$\sqrt{2+\sqrt{2+\sqrt{2+\dots\sqrt{2}}}} = 2 \cos \frac{\pi}{2^{k+2}}$$

From assumption add 2 to both sides and take square root to obtain $k+1$ lots of 2's on LHS.

$$\begin{aligned} \sqrt{2+\sqrt{2+\sqrt{2+\dots\sqrt{2}}}} &= \sqrt{2 + 2 \cos \frac{\pi}{2^{k+1}}} \\ &= \sqrt{2(1 + \cos \frac{\pi}{2^{k+1}})} \end{aligned}$$

$$\text{Let } \theta = \frac{\pi}{2^{k+1}}$$

$$\begin{aligned} \therefore \text{LHS} &= \sqrt{2(1 + \cos \theta)} \\ &= \sqrt{2} \sqrt{1 + \cos \theta} \\ &= \sqrt{2} \cdot \sqrt{2} \cos \frac{1}{2} \theta \\ &= 2 \cos \frac{1}{2} \cdot \frac{\pi}{2^{k+1}} \\ &= 2 \cos \frac{\pi}{2^{k+2}} \end{aligned}$$

= RHS $\therefore$ True for $n=k+1$.

$$\cos \theta = \sqrt{\frac{1 + \cos 2\theta}{2}}$$

$$\therefore \cos \frac{1}{2} \theta = \sqrt{\frac{1 + \cos \theta}{2}}$$

$$\sqrt{2} \cos \frac{1}{2} \theta = \sqrt{1 + \cos \theta}$$

Step 4: By Principle of Mathematical Induction true for all $n \geq 1$.